The stochastic $1/\log N$ regime

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Abstract

First unit of Astra #19's next programme, and its feasibility gate. For a corner-vanishing observable ($y_{\varphi,00} = 0$) with a random phase $X_n \Rightarrow Z$ in the coefficient-pair space, the chart posterior ratio decays like $1/\log N$ with a *random* limiting constant:

$$\log N \cdot \frac{Z_N[\varphi](X_n)}{Z_N[1](X_n)} \Rightarrow \frac{B_p^{\varphi}(Z)}{A_p^1(Z)}$$

(`tendstoInDistribution_log_posterior`). The gate passed with the existing infrastructure: the numerator's leading log coefficient vanishes identically for every phase, so its B-slot normalised remainder is exactly $Z_N[\varphi]/N^{-p}$; the B-slot error tends to zero in probability along the same random input as the denominator's A-slot error, giving joint convergence; the positive-denominator quotient theorem and the exact identity $\mathrm{normB}^{\varphi}/\mathrm{normA}^1 = (Z[\varphi]/Z[1])\log N$ finish. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem coeffA_withY_eq_zero (hy0 : y  (0, 0) = 0) (a) :
    coeffA    h  h  k  k  p (withY   h  y  hy  a) = 0
theorem normB_withY_eq (hy0) (N a) : normB   b   p p T h  h  k  k  p N (withY   h  y  hy  a)
    = chartZ   b   h  h  k  k  N (withY   h  y  hy  a) / N ^ (-p)
theorem normA_leading_eq (N a) : normA   b   p p T h  h  k  k  p N a
    = chartZ   b   h  h  k  k  N a / (N ^ (-p) * log N)
theorem normB_div_normA_eq (hy0) (hN : 1 < N) (a a') : normB … (withY y  a) / normA … a'
    = chartZ … (withY y  a) / chartZ … a' * log N
theorem tendstoInMeasure_normB_withY_sub ... : TendstoInMeasure   (fun n   =>
    normB … p (Nseq n) (withY y  (X n  )) - coeffB … p (withY y  (X n  ))) l (fun _ => 0)
theorem tendstoInDistribution_normBA_pair ... : TendstoInDistribution (fun n   =>
    (normB … (withY y  (X n  )), normA … (withY y  (X n  )))) l
    (fun   => (coeffB … (withY y  (Z  )), coeffA … (withY y  (Z  )))) (fun _ =>  )  '
theorem tendstoInDistribution_log_posterior ...
    (hy 0 : y  (0, 0) = 0) (hy 0 : 0 < y  (0, 0)) :
    TendstoInDistribution (fun n   => chartZ … (Nseq n) (withY y  (X n  ))
      / chartZ … (Nseq n) (withY y  (X n  )) * log (Nseq n)) l
    (fun   => coeffB … p (withY y  (Z  )) / coeffA … p (withY y  (Z  ))) (fun _ =>  )  '
```

2 Mathematics

Astra #19's gate obligations: (1) cancellation for the whole stochastic phase family — $A_p^{\varphi} = y_{\varphi,00} J_p(x_{00})/(k_1 k_2) = 0$ for every phase; (2) the correct scale of the numerator remainder — with

$a_N = N^{-p} \log N$ the B-slot theorem gives $Z_N[\varphi] - N^{-p} B_p^\varphi = o_{\mathbb{P}}(N^{-p}) = o_{\mathbb{P}}(a_N / \log N)$ exactly as required; (3) joint convergence from the common input — both errors are functions of X_n and the pair of limiting coefficients is a continuous image of Z; (4) almost-sure positivity of $A_p^1(Z)$ from $y_{1,00} > 0$. Everything reduces to units 148–159 (envelope bounds on balls, tightness, the approximation lemma, the clipped-division quotient theorem).

3 Paper-facing meaning

This is the stochastic form of the $1/\log N$ regime recorded in `rem:empirical_expectation_lean`: for an observable vanishing at the corner, the empirical posterior expectation decays like $1/\log N$ with a random coefficient $B_p^\varphi(Z)/A_p^1(Z)$ determined by the limiting coefficient array; when $B_p^\varphi(Z) = 0$ the limit is 0 and no faster rate is claimed.

4 Lean notes

The unit is a near-verbatim B-slot mirror of `FixedAmplitudeJointLimit.lean` and `ChartPosteriorLimit.lean`: `normB_sub_le`/`tendsto_errB` replace the A-slot bounds, `measurable_normB_comp` and `continuous_coeffB` need the extra hypothesis $p > 0$, and the final transfer uses `tendstoInMeasure_of_eventually_abs_le` with the exact identity for $N > 1$. It compiled at the first attempt.